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For the complex case

For the simple case

```
syms F q1 q2 q1d q2d lu lf real J = [0 0 lu*cos(q1)+lf*cos(q1+q2) lf*cos(q1+q2) lu*sin(q1)+lf*sin(q1+q2) lf*sin(q1+q2)];
```

```
v = J*[q1d;q2d]/norm(J*[q1d;q2d]); z2 = [1;0;0]; n = cross(z2,v); Fn = F*n; torq = simplify(J'*Fn)
```

Plot

```
% complex version lf = 1; lu = 1.2; q1 = 0; q2 = 1.2; q3 = 0; q4 = 0.5; q2d = 1; q4d = 2; q3d = 0; q1d = 0; F = 1; Ft = F; Fn = F; T01 = DHC(-pi/2, 0, 0, -pi/2 + q1); T12 = DHC(-pi/2, 0, 0, -pi/2 + q2); T23 = DHC(pi/2, 0, 0, q3); T33p = DHC(-pi/2, 0, 0, -pi/2); T3p4 = DHC(0, lu, 0, q4); T45 = DHC(0, lf, 0, 0); T02 = T01 * T12; T03 = T02 * T23;
```

```
T04 = T03 * T33p * T3p4;
```

```
T05 = T04 * T45; Pos = [zeros(3,1) T01(1:3,4) T02(1:3,4) T03(1:3,4) T04(1:3,4) T05(1:3,4)];
```

```
Jve = calculateJacobian(T01,T02,T03,T04,T05,5);
```

```
% for tangential motion ve = Jve *[q1d;q2d;q3d;q4d]/norm(Jve *[q1d;q2d;q3d;q4d]); z4 = T04(1:3,3)/norm(T04(1:3,3)); nt = cross(ve,z4); Fnt = Ft*nt; % force in the plane Fn_comp = Fnt; tangentialTorque = simplify(Jve'*Fnt) nn = cross(ve,nt); Fnn = Fn*nn; % force normal to the plane normalTorque = simplify(Jve'*Fnn) torq_comp = normalTorque;
```

```
figure() subplot(1,2,1) plot3(Pos(1,:),Pos(2,:),Pos(3:,:),'-o') hold on P = [Pos(:,end) Pos(:,end)+Fn]; plot3(P(1,:), P(2,:), P(3,:), 'LineWidth', 2); axis equal xlabel("x") ylabel("y") zlabel("z") axis([-5 5 -5 5 -5 5]) view(120,20)
```

```
% simple version q1 = q2; q2 = q4; q1d = q2d; q2d = q4d; J = [0 0 lu*cos(q1)+lf*cos(q1+q2) lf*cos(q1+q2) lu*sin(q1)+lf*sin(q1+q2) lf*sin(q1+q2)];
```

```
v = J*[q1d;q2d]/norm(J*[q1d;q2d]); pe = [0;lu*sin(q1)+lf*sin(q1+q2);-lu*cos(q1)-lf*cos(q1+q2)]/norm([0;lu*sin(q1)+lf*sin(q1+q2);-lu*cos(q1)-lf*cos(q1+q2)]); n = (pe-dot(pe,v)*v)/norm(pe-dot(pe,v)*v); % n = cross(v,[1;0;0]); Fn = F*n; torq = J'*Fn; Pos = [zeros(3,1) [0;lu*sin(q1);-lu*cos(q1)] [0;lu*sin(q1)+lf*sin(q1+q2);-lu*cos(q1)-lf*cos(q1+q2)]]; subplot(1,2,2) plot3(Pos(1,:),Pos(2,:),Pos(3:,:),'-o') hold on P = [Pos(:,end) Pos(:,end)+Fn]; plot3(P(1,:), P(2,:), P(3,:), 'LineWidth', 2); axis equal xlabel("x") ylabel("y") zlabel("z") axis([-5 5 -5 5 -5 5]) view(120,20) difference = norm(Fn_comp - Fn); fprintf('The difference between two reaction forces is %d\n', difference); difference = norm(torq_comp - [0; torq(1); 0 ;torq(2)]); fprintf('The difference between two torques is %d\n', difference);
```

Functions

tangentialTorque =

$$\begin{aligned} & -(Ft*lf*lu*q4d*cos(q2)*sin(q3)*sin(q4))/ \\ & ((lu*q2d*cos(q2) - lf*q4d*sin(q2)*sin(q4) + \\ & lf*q2d*cos(q2)*cos(q4) + lf*q4d*cos(q2)*cos(q3)*cos(q4) - \\ & lf*q2d*cos(q3)*sin(q2)*sin(q4) - lf*q3d*cos(q2)*sin(q3)*sin(q4))^2 \\ & + (lu*q1d*cos(q2)*sin(q1) + lu*q2d*cos(q1)*sin(q2) + \\ & lf*q1d*cos(q2)*cos(q4)*sin(q1) + lf*q2d*cos(q1)*cos(q4)*sin(q2) \\ & + lf*q4d*cos(q1)*cos(q2)*sin(q4) - lf*q1d*cos(q1)*sin(q3)*sin(q4) \\ & - lf*q3d*cos(q3)*sin(q1)*sin(q4) - lf*q4d*cos(q4)*sin(q1)*sin(q3) \\ & + lf*q2d*cos(q1)*cos(q2)*cos(q3)*sin(q4) \\ & + lf*q4d*cos(q1)*cos(q3)*cos(q4)*sin(q2) \\ & - lf*q1d*cos(q3)*sin(q1)*sin(q2)*sin(q4) - \\ & lf*q3d*cos(q1)*sin(q2)*sin(q3)*sin(q4))^2 + \\ & (lu*q2d*sin(q1)*sin(q2) - lu*q1d*cos(q1)*cos(q2) - \\ & lf*q1d*cos(q1)*cos(q2)*cos(q4) + lf*q3d*cos(q1)*cos(q3)*sin(q4) \\ & + lf*q4d*cos(q1)*cos(q4)*sin(q3) + lf*q2d*cos(q4)*sin(q1)*sin(q2) \\ & + lf*q4d*cos(q2)*sin(q1)*sin(q4) - lf*q1d*sin(q1)*sin(q3)*sin(q4) \\ & + lf*q1d*cos(q1)*cos(q3)*sin(q2)*sin(q4) + \\ & lf*q2d*cos(q2)*cos(q3)*sin(q1)*sin(q4) + \\ & lf*q4d*cos(q3)*cos(q4)*sin(q1)*sin(q2) - \\ & lf*q3d*sin(q1)*sin(q2)*sin(q3)*sin(q4))^2)^{(1/2)} \\ & (Ft*lf*lu*q4d*cos(q3)*sin(q4))/ \\ & ((lu*q2d*cos(q2) - lf*q4d*sin(q2)*sin(q4) + \\ & lf*q2d*cos(q2)*cos(q4) + lf*q4d*cos(q2)*cos(q3)*cos(q4) - \\ & lf*q2d*cos(q3)*sin(q2)*sin(q4) - lf*q3d*cos(q2)*sin(q3)*sin(q4))^2 \\ & + (lu*q1d*cos(q2)*sin(q1) + lu*q2d*cos(q1)*sin(q2) + \\ & lf*q1d*cos(q2)*cos(q4)*sin(q1) + lf*q2d*cos(q1)*cos(q4)*sin(q2) \\ & + lf*q4d*cos(q1)*cos(q2)*sin(q4) - lf*q1d*cos(q1)*sin(q3)*sin(q4) \\ & - lf*q3d*cos(q3)*sin(q1)*sin(q4) - lf*q4d*cos(q4)*sin(q1)*sin(q3) \\ & + lf*q2d*cos(q1)*cos(q2)*cos(q3)*sin(q4) \\ & + lf*q4d*cos(q1)*cos(q3)*cos(q4)*sin(q2) \\ & - lf*q1d*cos(q3)*sin(q1)*sin(q2)*sin(q4) - \\ & lf*q3d*cos(q1)*sin(q2)*sin(q3)*sin(q4))^2 + \\ & (lu*q2d*sin(q1)*sin(q2) - lu*q1d*cos(q1)*cos(q2) - \\ & lf*q1d*cos(q1)*cos(q2)*cos(q4) + lf*q3d*cos(q1)*cos(q3)*sin(q4) \\ & + lf*q4d*cos(q1)*cos(q4)*sin(q3) + lf*q2d*cos(q4)*sin(q1)*sin(q2) \\ & + lf*q4d*cos(q2)*sin(q1)*sin(q4) - lf*q1d*sin(q1)*sin(q3)*sin(q4) \\ & + lf*q1d*cos(q1)*cos(q3)*sin(q2)*sin(q4) + \\ & lf*q2d*cos(q2)*cos(q3)*sin(q1)*sin(q4) + \\ & lf*q4d*cos(q3)*cos(q4)*sin(q1)*sin(q2) - \\ & lf*q3d*sin(q1)*sin(q2)*sin(q3)*sin(q4))^2)^{(1/2)} \end{aligned}$$

$$\begin{aligned}
& 0 \\
& -(F_t * l_f * l_u * \sin(q_4) * (q_{2d} * \cos(q_3) - q_{1d} * \cos(q_2) * \sin(q_3))) / \\
& ((l_u * q_{2d} * \cos(q_2) - l_f * q_{4d} * \sin(q_2) * \sin(q_4) + \\
& l_f * q_{2d} * \cos(q_2) * \cos(q_4) + l_f * q_{4d} * \cos(q_2) * \cos(q_3) * \cos(q_4) - \\
& l_f * q_{2d} * \cos(q_3) * \sin(q_2) * \sin(q_4) - l_f * q_{3d} * \cos(q_2) * \sin(q_3) * \sin(q_4)) ^2 \\
& + (l_u * q_{1d} * \cos(q_2) * \sin(q_1) + l_u * q_{2d} * \cos(q_1) * \sin(q_2) + \\
& l_f * q_{1d} * \cos(q_2) * \cos(q_4) * \sin(q_1) + l_f * q_{2d} * \cos(q_1) * \cos(q_4) * \sin(q_2) \\
& + l_f * q_{4d} * \cos(q_1) * \cos(q_2) * \sin(q_4) - l_f * q_{1d} * \cos(q_1) * \sin(q_3) * \sin(q_4) \\
& - l_f * q_{3d} * \cos(q_3) * \sin(q_1) * \sin(q_4) - l_f * q_{4d} * \cos(q_4) * \sin(q_1) * \sin(q_3) \\
& + l_f * q_{2d} * \cos(q_1) * \cos(q_2) * \cos(q_3) * \sin(q_4) \\
& + l_f * q_{4d} * \cos(q_1) * \cos(q_3) * \cos(q_4) * \sin(q_2) \\
& - l_f * q_{1d} * \cos(q_3) * \sin(q_1) * \sin(q_2) * \sin(q_4) - \\
& l_f * q_{3d} * \cos(q_1) * \sin(q_2) * \sin(q_3) * \sin(q_4)) ^2 + \\
& (l_u * q_{2d} * \sin(q_1) * \sin(q_2) - l_u * q_{1d} * \cos(q_1) * \cos(q_2) - \\
& l_f * q_{1d} * \cos(q_1) * \cos(q_2) * \cos(q_4) + l_f * q_{3d} * \cos(q_1) * \cos(q_3) * \sin(q_4) \\
& + l_f * q_{4d} * \cos(q_1) * \cos(q_4) * \sin(q_3) + l_f * q_{2d} * \cos(q_4) * \sin(q_1) * \sin(q_2) \\
& + l_f * q_{4d} * \cos(q_2) * \sin(q_1) * \sin(q_4) - l_f * q_{1d} * \sin(q_1) * \sin(q_3) * \sin(q_4) \\
& + l_f * q_{1d} * \cos(q_1) * \cos(q_3) * \sin(q_2) * \sin(q_4) + \\
& l_f * q_{2d} * \cos(q_2) * \cos(q_3) * \sin(q_1) * \sin(q_4) + \\
& l_f * q_{4d} * \cos(q_3) * \cos(q_4) * \sin(q_1) * \sin(q_2) - \\
& l_f * q_{3d} * \sin(q_1) * \sin(q_2) * \sin(q_3) * \sin(q_4)) ^2) ^{(1/2)}
\end{aligned}$$

normalTorque =

$$\begin{aligned}
& \text{Fn}(((\text{lu} * \text{q2d} * \cos(\text{q2}) \\
& - \text{lf} * \text{q4d} * \sin(\text{q2}) * \sin(\text{q4}) + \text{lf} * \text{q2d} * \cos(\text{q2}) * \cos(\text{q4}) + \\
& \text{lf} * \text{q4d} * \cos(\text{q2}) * \cos(\text{q3}) * \cos(\text{q4}) - \text{lf} * \text{q2d} * \cos(\text{q3}) * \sin(\text{q2}) * \sin(\text{q4}) - \\
& \text{lf} * \text{q3d} * \cos(\text{q2}) * \sin(\text{q3}) * \sin(\text{q4})) * (\text{lf} * \text{q4d} * \cos(\text{q1}) * \cos(\text{q2}) * \cos(\text{q4}) - \\
& \text{lu} * \text{q1d} * \cos(\text{q1}) * \cos(\text{q2})^2 * \sin(\text{q3}) + \text{lu} * \text{q2d} * \cos(\text{q1}) * \cos(\text{q2}) * \cos(\text{q3}) \\
& - \text{lf} * \text{q1d} * \cos(\text{q2}) * \sin(\text{q1}) * \sin(\text{q4}) + \text{lf} * \text{q4d} * \sin(\text{q1}) * \sin(\text{q3}) * \sin(\text{q4}) \\
& + \text{lf} * \text{q1d} * \cos(\text{q2}) * \cos(\text{q3})^2 * \sin(\text{q1}) * \sin(\text{q4}) \\
& - \text{lf} * \text{q2d} * \cos(\text{q1}) * \cos(\text{q3})^2 * \sin(\text{q2}) * \sin(\text{q4}) \\
& + \text{lf} * \text{q2d} * \cos(\text{q1}) * \cos(\text{q2}) * \cos(\text{q3}) * \cos(\text{q4}) \\
& - \text{lf} * \text{q4d} * \cos(\text{q1}) * \cos(\text{q3}) * \sin(\text{q2}) * \sin(\text{q4}) \\
& + \text{lf} * \text{q2d} * \cos(\text{q3}) * \sin(\text{q1}) * \sin(\text{q3}) * \sin(\text{q4}) - \\
& \text{lf} * \text{q1d} * \cos(\text{q1}) * \cos(\text{q2})^2 * \cos(\text{q4}) * \sin(\text{q3}) + \\
& \text{lf} * \text{q1d} * \cos(\text{q1}) * \cos(\text{q2}) * \cos(\text{q3}) * \sin(\text{q2}) * \sin(\text{q3}) * \sin(\text{q4}))) / \\
& ((\text{lu} * \text{q2d} * \cos(\text{q2}) - \text{lf} * \text{q4d} * \sin(\text{q2}) * \sin(\text{q4}) + \text{lf} * \text{q2d} * \cos(\text{q2}) * \cos(\text{q4}) \\
& + \text{lf} * \text{q4d} * \cos(\text{q2}) * \cos(\text{q3}) * \cos(\text{q4}) - \text{lf} * \text{q2d} * \cos(\text{q3}) * \sin(\text{q2}) * \sin(\text{q4}) \\
& - \text{lf} * \text{q3d} * \cos(\text{q2}) * \sin(\text{q3}) * \sin(\text{q4}))^2 + (\text{q4d} * (\text{lf} * (\cos(\text{q4}) * \sin(\text{q2}) + \\
& \cos(\text{q2}) * \cos(\text{q3}) * \sin(\text{q4})) * (\cos(\text{q1}) * \cos(\text{q3}) - \sin(\text{q1}) * \sin(\text{q2}) * \sin(\text{q3})) \\
& + \text{lf} * \cos(\text{q2}) * \sin(\text{q3}) * (\cos(\text{q1}) * \sin(\text{q3}) * \sin(\text{q4}) - \\
& \cos(\text{q2}) * \cos(\text{q4}) * \sin(\text{q1}) + \cos(\text{q3}) * \sin(\text{q1}) * \sin(\text{q2}) * \sin(\text{q4}))) \\
& + \text{q1d} * (\text{lu} * \cos(\text{q2}) * \sin(\text{q1}) + \text{lf} * \cos(\text{q2}) * \cos(\text{q4}) * \sin(\text{q1}) - \\
& \text{lf} * \cos(\text{q1}) * \sin(\text{q3}) * \sin(\text{q4}) - \text{lf} * \cos(\text{q3}) * \sin(\text{q1}) * \sin(\text{q2}) * \sin(\text{q4})) \\
& + \text{q2d} * \cos(\text{q1}) * (\text{lu} * \sin(\text{q2}) + \text{lf} * \cos(\text{q4}) * \sin(\text{q2}) + \\
& \text{lf} * \cos(\text{q2}) * \cos(\text{q3}) * \sin(\text{q4})) - \text{lf} * \text{q3d} * \sin(\text{q4}) * (\cos(\text{q3}) * \sin(\text{q1}) + \\
& \cos(\text{q1}) * \sin(\text{q2}) * \sin(\text{q3})))^2 + (\text{lf} * \text{q4d} * (\cos(\text{q1}) * \cos(\text{q4}) * \sin(\text{q3}) \\
& + \cos(\text{q2}) * \sin(\text{q1}) * \sin(\text{q4}) + \cos(\text{q3}) * \cos(\text{q4}) * \sin(\text{q1}) * \sin(\text{q2})) \\
& - \text{q1d} * (\text{lu} * \cos(\text{q1}) * \cos(\text{q2}) + \text{lf} * \cos(\text{q1}) * \cos(\text{q2}) * \cos(\text{q4}) + \\
& \text{lf} * \sin(\text{q1}) * \sin(\text{q3}) * \sin(\text{q4}) - \text{lf} * \cos(\text{q1}) * \cos(\text{q3}) * \sin(\text{q2}) * \sin(\text{q4})) \\
& + \text{q2d} * \sin(\text{q1}) * (\text{lu} * \sin(\text{q2}) + \text{lf} * \cos(\text{q4}) * \sin(\text{q2}) + \\
& \text{lf} * \cos(\text{q2}) * \cos(\text{q3}) * \sin(\text{q4})) + \text{lf} * \text{q3d} * \sin(\text{q4}) * (\cos(\text{q1}) * \cos(\text{q3}) \\
& - \sin(\text{q1}) * \sin(\text{q2}) * \sin(\text{q3})))^2) + ((\text{q4d} * (\text{lf} * (\cos(\text{q4}) * \sin(\text{q2}) +
\end{aligned}$$

$$\begin{aligned}
& \cos(q_2) \cos(q_3) \sin(q_4)) (\cos(q_1) \cos(q_3) - \sin(q_1) \sin(q_2) \sin(q_3)) \\
& + lf \cos(q_2) \sin(q_3) (\cos(q_1) \sin(q_3) \sin(q_4) - \\
& \cos(q_2) \cos(q_4) \sin(q_1) + \cos(q_3) \sin(q_1) \sin(q_2) \sin(q_4))) \\
& + q1d (lu \cos(q_2) \sin(q_1) + lf \cos(q_2) \cos(q_4) \sin(q_1) - \\
& lf \cos(q_1) \sin(q_3) \sin(q_4) - lf \cos(q_3) \sin(q_1) \sin(q_2) \sin(q_4)) \\
& + q2d \cos(q_1) (lu \sin(q_2) + lf \cos(q_4) \sin(q_2) + \\
& lf \cos(q_2) \cos(q_3) \sin(q_4)) - lf q3d \sin(q_4) (\cos(q_3) \sin(q_1) \\
& + \cos(q_1) \sin(q_2) \sin(q_3))) (lf q4d \cos(q_4) \sin(q_2) + \\
& lu q2d \cos(q_3) \sin(q_2) + lf q2d \cos(q_2) \cos(q_3)^2 \sin(q_4) + \\
& lf q2d \cos(q_3) \cos(q_4) \sin(q_2) + lf q4d \cos(q_2) \cos(q_3) \sin(q_4) - \\
& lu q1d \cos(q_2) \sin(q_2) \sin(q_3) - \\
& lf q1d \cos(q_2)^2 \cos(q_3) \sin(q_3) \sin(q_4) - \\
& lf q1d \cos(q_2) \cos(q_4) \sin(q_2) \sin(q_3))) / ((lu q2d \cos(q_2) \\
& - lf q4d \sin(q_2) \sin(q_4) + lf q2d \cos(q_2) \cos(q_4) + \\
& lf q4d \cos(q_2) \cos(q_3) \cos(q_4) - lf q2d \cos(q_3) \sin(q_2) \sin(q_4) \\
& - lf q3d \cos(q_2) \sin(q_3) \sin(q_4))^2 + (q4d (lf (\cos(q_4) \sin(q_2) + \\
& \cos(q_2) \cos(q_3) \sin(q_4)) (\cos(q_1) \cos(q_3) - \sin(q_1) \sin(q_2) \sin(q_3)) \\
& + lf \cos(q_2) \sin(q_3) (\cos(q_1) \sin(q_3) \sin(q_4) - \\
& \cos(q_2) \cos(q_4) \sin(q_1) + \cos(q_3) \sin(q_1) \sin(q_2) \sin(q_4))) \\
& + q1d (lu \cos(q_2) \sin(q_1) + lf \cos(q_2) \cos(q_4) \sin(q_1) - \\
& lf \cos(q_1) \sin(q_3) \sin(q_4) - lf \cos(q_3) \sin(q_1) \sin(q_2) \sin(q_4)) \\
& + q2d \cos(q_1) (lu \sin(q_2) + lf \cos(q_4) \sin(q_2) + \\
& lf \cos(q_2) \cos(q_3) \sin(q_4)) - lf q3d \sin(q_4) (\cos(q_3) \sin(q_1) + \\
& \cos(q_1) \sin(q_2) \sin(q_3)))^2 + (lf q4d (\cos(q_1) \cos(q_4) \sin(q_3) \\
& + \cos(q_2) \sin(q_1) \sin(q_4) + \cos(q_3) \cos(q_4) \sin(q_1) \sin(q_2)) \\
& - q1d (lu \cos(q_1) \cos(q_2) + lf \cos(q_1) \cos(q_2) \cos(q_4) + \\
& lf \sin(q_1) \sin(q_3) \sin(q_4) - lf \cos(q_1) \cos(q_3) \sin(q_2) \sin(q_4)) \\
& + q2d \sin(q_1) (lu \sin(q_2) + lf \cos(q_4) \sin(q_2) + \\
& lf \cos(q_2) \cos(q_3) \sin(q_4)) + lf q3d \sin(q_4) (\cos(q_1) \cos(q_3) \\
& - \sin(q_1) \sin(q_2) \sin(q_3)))^2)) (lu \cos(q_1) \cos(q_2) + \\
& lf \cos(q_1) \cos(q_2) \cos(q_4) + lf \sin(q_1) \sin(q_3) \sin(q_4) \\
& - lf \cos(q_1) \cos(q_3) \sin(q_2) \sin(q_4)) + \\
& Fn (((lf q4d (\cos(q_1) \cos(q_4) \sin(q_3) + \\
& \cos(q_2) \sin(q_1) \sin(q_4) + \cos(q_3) \cos(q_4) \sin(q_1) \sin(q_2)) \\
& - q1d (lu \cos(q_1) \cos(q_2) + lf \cos(q_1) \cos(q_2) \cos(q_4) + \\
& lf \sin(q_1) \sin(q_3) \sin(q_4) - lf \cos(q_1) \cos(q_3) \sin(q_2) \sin(q_4)) \\
& + q2d \sin(q_1) (lu \sin(q_2) + lf \cos(q_4) \sin(q_2) + \\
& lf \cos(q_2) \cos(q_3) \sin(q_4)) + lf q3d \sin(q_4) (\cos(q_1) \cos(q_3) \\
& - \sin(q_1) \sin(q_2) \sin(q_3))) (lf q4d \cos(q_4) \sin(q_2) + \\
& lu q2d \cos(q_3) \sin(q_2) + lf q2d \cos(q_2) \cos(q_3)^2 \sin(q_4) + \\
& lf q2d \cos(q_3) \cos(q_4) \sin(q_2) + lf q4d \cos(q_2) \cos(q_3) \sin(q_4) - \\
& lu q1d \cos(q_2) \sin(q_2) \sin(q_3) - \\
& lf q1d \cos(q_2)^2 \cos(q_3) \sin(q_3) \sin(q_4) - \\
& lf q1d \cos(q_2) \cos(q_4) \sin(q_2) \sin(q_3))) / ((lu q2d \cos(q_2) \\
& - lf q4d \sin(q_2) \sin(q_4) + lf q2d \cos(q_2) \cos(q_4) + \\
& lf q4d \cos(q_2) \cos(q_3) \cos(q_4) - lf q2d \cos(q_3) \sin(q_2) \sin(q_4) \\
& - lf q3d \cos(q_2) \sin(q_3) \sin(q_4))^2 + (q4d (lf (\cos(q_4) \sin(q_2) + \\
& \cos(q_2) \cos(q_3) \sin(q_4)) (\cos(q_1) \cos(q_3) - \sin(q_1) \sin(q_2) \sin(q_3)) \\
& + lf \cos(q_2) \sin(q_3) (\cos(q_1) \sin(q_3) \sin(q_4) - \\
& \cos(q_2) \cos(q_4) \sin(q_1) + \cos(q_3) \sin(q_1) \sin(q_2) \sin(q_4))) \\
& + q1d (lu \cos(q_2) \sin(q_1) + lf \cos(q_2) \cos(q_4) \sin(q_1) - \\
& lf \cos(q_1) \sin(q_3) \sin(q_4) - lf \cos(q_3) \sin(q_1) \sin(q_2) \sin(q_4)) \\
& + q2d \cos(q_1) (lu \sin(q_2) + lf \cos(q_4) \sin(q_2) + \\
& lf \cos(q_2) \cos(q_3) \sin(q_4)) - lf q3d \sin(q_4) (\cos(q_3) \sin(q_1) +
\end{aligned}$$

$$\begin{aligned}
& \cos(q_1) \sin(q_2) \sin(q_3)) \wedge^2 + (lf \cdot q_{4d} (\cos(q_1) \cos(q_4) \sin(q_3) \\
& + \cos(q_2) \sin(q_1) \sin(q_4) + \cos(q_3) \cos(q_4) \sin(q_1) \sin(q_2)) \\
& - q_{1d} (lu \cos(q_1) \cos(q_2) + lf \cos(q_1) \cos(q_2) \cos(q_4) + \\
& lf \sin(q_1) \sin(q_3) \sin(q_4) - lf \cos(q_1) \cos(q_3) \sin(q_2) \sin(q_4)) \\
& + q_{2d} \sin(q_1) (lu \sin(q_2) + lf \cos(q_4) \sin(q_2) + \\
& lf \cos(q_2) \cos(q_3) \sin(q_4)) + lf \cdot q_{3d} \sin(q_4) (\cos(q_1) \cos(q_3) \\
& - \sin(q_1) \sin(q_2) \sin(q_3)) \wedge^2) - ((lu \cdot q_{2d} \cos(q_2) \\
& - lf \cdot q_{4d} \sin(q_2) \sin(q_4) + lf \cdot q_{2d} \cos(q_2) \cos(q_4) + \\
& lf \cdot q_{4d} \cos(q_2) \cos(q_3) \cos(q_4) - lf \cdot q_{2d} \cos(q_3) \sin(q_2) \sin(q_4) - \\
& lf \cdot q_{3d} \cos(q_2) \sin(q_3) \sin(q_4)) (lu \cdot q_{1d} \cos(q_2) \wedge^2 \sin(q_1) \sin(q_3) \\
& - lf \cdot q_{1d} \cos(q_1) \cos(q_2) \sin(q_4) - lf \cdot q_{4d} \cos(q_2) \cos(q_4) \sin(q_1) \\
& - lu \cdot q_{2d} \cos(q_2) \cos(q_3) \sin(q_1) + lf \cdot q_{4d} \cos(q_1) \sin(q_3) \sin(q_4) \\
& + lf \cdot q_{1d} \cos(q_2) \wedge^2 \cos(q_4) \sin(q_1) \sin(q_3) \\
& + lf \cdot q_{2d} \cos(q_3) \wedge^2 \sin(q_1) \sin(q_2) \sin(q_4) \\
& - lf \cdot q_{2d} \cos(q_2) \cos(q_3) \cos(q_4) \sin(q_1) \\
& + lf \cdot q_{2d} \cos(q_1) \cos(q_3) \sin(q_3) \sin(q_4) \\
& + lf \cdot q_{4d} \cos(q_3) \sin(q_1) \sin(q_2) \sin(q_4) + \\
& lf \cdot q_{1d} \cos(q_1) \cos(q_2) \cos(q_3) \wedge^2 \sin(q_4) - \\
& lf \cdot q_{1d} \cos(q_2) \cos(q_3) \sin(q_1) \sin(q_2) \sin(q_3) \sin(q_4))) / \\
& ((lu \cdot q_{2d} \cos(q_2) - lf \cdot q_{4d} \sin(q_2) \sin(q_4) + lf \cdot q_{2d} \cos(q_2) \cos(q_4) \\
& + lf \cdot q_{4d} \cos(q_2) \cos(q_3) \cos(q_4) - lf \cdot q_{2d} \cos(q_3) \sin(q_2) \sin(q_4) \\
& - lf \cdot q_{3d} \cos(q_2) \sin(q_3) \sin(q_4)) \wedge^2 + (q_{4d} (lf (\cos(q_4) \sin(q_2) + \\
& \cos(q_2) \cos(q_3) \sin(q_4)) (\cos(q_1) \cos(q_3) - \sin(q_1) \sin(q_2) \sin(q_3)) \\
& + lf \cos(q_2) \sin(q_3) (\cos(q_1) \sin(q_3) \sin(q_4) - \\
& \cos(q_2) \cos(q_4) \sin(q_1) + \cos(q_3) \sin(q_1) \sin(q_2) \sin(q_4))) \\
& + q_{1d} (lu \cos(q_2) \sin(q_1) + lf \cos(q_2) \cos(q_4) \sin(q_1) - \\
& lf \cos(q_1) \sin(q_3) \sin(q_4) - lf \cos(q_3) \sin(q_1) \sin(q_2) \sin(q_4)) \\
& + q_{2d} \cos(q_1) (lu \sin(q_2) + lf \cos(q_4) \sin(q_2) + \\
& lf \cos(q_2) \cos(q_3) \sin(q_4)) - lf \cdot q_{3d} \sin(q_4) (\cos(q_3) \sin(q_1) + \\
& \cos(q_1) \sin(q_2) \sin(q_3)) \wedge^2 + (lf \cdot q_{4d} (\cos(q_1) \cos(q_4) \sin(q_3) \\
& + \cos(q_2) \sin(q_1) \sin(q_4) + \cos(q_3) \cos(q_4) \sin(q_1) \sin(q_2)) \\
& - q_{1d} (lu \cos(q_1) \cos(q_2) + lf \cos(q_1) \cos(q_2) \cos(q_4) + \\
& lf \sin(q_1) \sin(q_3) \sin(q_4) - lf \cos(q_1) \cos(q_3) \sin(q_2) \sin(q_4)) \\
& + q_{2d} \sin(q_1) (lu \sin(q_2) + lf \cos(q_4) \sin(q_2) + \\
& lf \cos(q_2) \cos(q_3) \sin(q_4)) + lf \cdot q_{3d} \sin(q_4) (\cos(q_1) \cos(q_3) \\
& - \sin(q_1) \sin(q_2) \sin(q_3)) \wedge^2)) (lu \cos(q_2) \sin(q_1) + \\
& lf \cos(q_2) \cos(q_4) \sin(q_1) - lf \cos(q_1) \sin(q_3) \sin(q_4) - \\
& lf \cos(q_3) \sin(q_1) \sin(q_2) \sin(q_4))
\end{aligned}$$

$$\begin{aligned}
& Fn \cos(q_1) (((lf \cdot q_{4d} (\cos(q_1) \cos(q_4) \sin(q_3) + \\
& \cos(q_2) \sin(q_1) \sin(q_4) + \cos(q_3) \cos(q_4) \sin(q_1) \sin(q_2)) \\
& - q_{1d} (lu \cos(q_1) \cos(q_2) + lf \cos(q_1) \cos(q_2) \cos(q_4) + \\
& lf \sin(q_1) \sin(q_3) \sin(q_4) - lf \cos(q_1) \cos(q_3) \sin(q_2) \sin(q_4)) \\
& + q_{2d} \sin(q_1) (lu \sin(q_2) + lf \cos(q_4) \sin(q_2) + \\
& lf \cos(q_2) \cos(q_3) \sin(q_4)) + lf \cdot q_{3d} \sin(q_4) (\cos(q_1) \cos(q_3) \\
& - \sin(q_1) \sin(q_2) \sin(q_3))) (lf \cdot q_{4d} \cos(q_4) \sin(q_2) + \\
& lu \cdot q_{2d} \cos(q_3) \sin(q_2) + lf \cdot q_{2d} \cos(q_2) \cos(q_3) \wedge^2 \sin(q_4) + \\
& lf \cdot q_{2d} \cos(q_3) \cos(q_4) \sin(q_2) + lf \cdot q_{4d} \cos(q_2) \cos(q_3) \sin(q_4) - \\
& lu \cdot q_{1d} \cos(q_2) \sin(q_2) \sin(q_3) - \\
& lf \cdot q_{1d} \cos(q_2) \wedge^2 \cos(q_3) \sin(q_3) \sin(q_4) - \\
& lf \cdot q_{1d} \cos(q_2) \cos(q_4) \sin(q_2) \sin(q_3))) / ((lu \cdot q_{2d} \cos(q_2) \\
& - lf \cdot q_{4d} \sin(q_2) \sin(q_4) + lf \cdot q_{2d} \cos(q_2) \cos(q_4) + \\
& lf \cdot q_{4d} \cos(q_2) \cos(q_3) \cos(q_4) - lf \cdot q_{2d} \cos(q_3) \sin(q_2) \sin(q_4)
\end{aligned}$$

$$\begin{aligned}
& - lf*q3d*cos(q2)*sin(q3)*sin(q4))^2 + (q4d*(lf*(cos(q4)*sin(q2) + \\
& cos(q2)*cos(q3)*sin(q4))*(cos(q1)*cos(q3) - sin(q1)*sin(q2)*sin(q3)) \\
& + lf*cos(q2)*sin(q3)*(cos(q1)*sin(q3)*sin(q4) - \\
& cos(q2)*cos(q4)*sin(q1) + cos(q3)*sin(q1)*sin(q2)*sin(q4))) \\
& + q1d*(lu*cos(q2)*sin(q1) + lf*cos(q2)*cos(q4)*sin(q1) - \\
& lf*cos(q1)*sin(q3)*sin(q4) - lf*cos(q3)*sin(q1)*sin(q2)*sin(q4)) \\
& + q2d*cos(q1)*(lu*sin(q2) + lf*cos(q4)*sin(q2) + \\
& lf*cos(q2)*cos(q3)*sin(q4)) - lf*q3d*sin(q4)*(cos(q3)*sin(q1) + \\
& cos(q1)*sin(q2)*sin(q3)))^2 + (lf*q4d*(cos(q1)*cos(q4)*sin(q3) \\
& + cos(q2)*sin(q1)*sin(q4) + cos(q3)*cos(q4)*sin(q1)*sin(q2)) \\
& - q1d*(lu*cos(q1)*cos(q2) + lf*cos(q1)*cos(q2)*cos(q4) + \\
& lf*sin(q1)*sin(q3)*sin(q4) - lf*cos(q1)*cos(q3)*sin(q2)*sin(q4)) \\
& + q2d*sin(q1)*(lu*sin(q2) + lf*cos(q4)*sin(q2) + \\
& lf*cos(q2)*cos(q3)*sin(q4)) + lf*q3d*sin(q4)*(cos(q1)*cos(q3) \\
& - sin(q1)*sin(q2)*sin(q3)))^2) - ((lu*q2d*cos(q2) \\
& - lf*q4d*sin(q2)*sin(q4) + lf*q2d*cos(q2)*cos(q4) + \\
& lf*q4d*cos(q2)*cos(q3)*cos(q4) - lf*q2d*cos(q3)*sin(q2)*sin(q4) - \\
& lf*q3d*cos(q2)*sin(q3)*sin(q4))*(lu*q1d*cos(q2)^2*sin(q1)*sin(q3) \\
& - lf*q1d*cos(q1)*cos(q2)*sin(q4) - lf*q4d*cos(q2)*cos(q4)*sin(q1) \\
& - lu*q2d*cos(q2)*cos(q3)*sin(q1) + lf*q4d*cos(q1)*sin(q3)*sin(q4) \\
& + lf*q1d*cos(q2)^2*cos(q4)*sin(q1)*sin(q3) \\
& + lf*q2d*cos(q3)^2*sin(q1)*sin(q2)*sin(q4) \\
& - lf*q2d*cos(q2)*cos(q3)*cos(q4)*sin(q1) \\
& + lf*q2d*cos(q1)*cos(q3)*sin(q3)*sin(q4) \\
& + lf*q4d*cos(q3)*sin(q1)*sin(q2)*sin(q4) + \\
& lf*q1d*cos(q1)*cos(q2)*cos(q3)^2*sin(q4) - \\
& lf*q1d*cos(q2)*cos(q3)*sin(q1)*sin(q2)*sin(q3)*sin(q4)))/ \\
& ((lu*q2d*cos(q2) - lf*q4d*sin(q2)*sin(q4) + lf*q2d*cos(q2)*cos(q4) \\
& + lf*q4d*cos(q2)*cos(q3)*cos(q4) - lf*q2d*cos(q3)*sin(q2)*sin(q4) \\
& - lf*q3d*cos(q2)*sin(q3)*sin(q4))^2 + (q4d*(lf*(cos(q4)*sin(q2) + \\
& cos(q2)*cos(q3)*sin(q4))*(cos(q1)*cos(q3) - sin(q1)*sin(q2)*sin(q3)) \\
& + lf*cos(q2)*sin(q3)*(cos(q1)*sin(q3)*sin(q4) - \\
& cos(q2)*cos(q4)*sin(q1) + cos(q3)*sin(q1)*sin(q2)*sin(q4))) \\
& + q1d*(lu*cos(q2)*sin(q1) + lf*cos(q2)*cos(q4)*sin(q1) - \\
& lf*cos(q1)*sin(q3)*sin(q4) - lf*cos(q3)*sin(q1)*sin(q2)*sin(q4)) \\
& + q2d*cos(q1)*(lu*sin(q2) + lf*cos(q4)*sin(q2) + \\
& lf*cos(q2)*cos(q3)*sin(q4)) - lf*q3d*sin(q4)*(cos(q3)*sin(q1) + \\
& cos(q1)*sin(q2)*sin(q3)))^2 + (lf*q4d*(cos(q1)*cos(q4)*sin(q3) \\
& + cos(q2)*sin(q1)*sin(q4) + cos(q3)*cos(q4)*sin(q1)*sin(q2)) \\
& - q1d*(lu*cos(q1)*cos(q2) + lf*cos(q1)*cos(q2)*cos(q4) + \\
& lf*sin(q1)*sin(q3)*sin(q4) - lf*cos(q1)*cos(q3)*sin(q2)*sin(q4)) \\
& + q2d*sin(q1)*(lu*sin(q2) + lf*cos(q4)*sin(q2) + \\
& lf*cos(q2)*cos(q3)*sin(q4)) + lf*q3d*sin(q4)*(cos(q1)*cos(q3) - \\
& sin(q1)*sin(q2)*sin(q3)))^2)*(lu*sin(q2) + lf*cos(q4)*sin(q2) + \\
& lf*cos(q2)*cos(q3)*sin(q4)) + (Fn*(lu*cos(q2) + lf*cos(q2)*cos(q4) \\
& - lf*cos(q3)*sin(q2)*sin(q4))*(lu^2*q1d^2*cos(q2)^3*sin(q3) - \\
& lf^2*q1d*q4d - lf^2*q1d*q4d*cos(q2)^2 + lf^2*q1d*q4d*cos(q4)^2 \\
& + lf^2*q1d^2*cos(q2)*sin(q3) + lf^2*q4d^2*cos(q2)*sin(q3) \\
& - lf^2*q1d*q2d*cos(q3) - lf^2*q3d*q4d*sin(q2) - \\
& lf^2*q1d*q2d*cos(q2)^2*cos(q3) + lf^2*q1d*q2d*cos(q3)*cos(q4)^2 \\
& - lu^2*q1d*q2d*cos(q2)^2*cos(q3) + lf^2*q3d*q4d*cos(q4)^2*sin(q2) \\
& - lf^2*q1d^2*cos(q2)^3*cos(q3)^2*sin(q3) \\
& + lf^2*q1d^2*cos(q2)^3*cos(q4)^2*sin(q3) \\
& + 2*lf^2*q1d*q2d*cos(q2)^2*cos(q3)^3 +
\end{aligned}$$

$$\begin{aligned}
& 2*lf^2*q1d*q4d*cos(q2)^2*cos(q3)^2 - lf^2*q1d*q4d*cos(q2)^2*cos(q4)^2 \\
& - lf^2*q2d*q3d*cos(q3)*sin(q2) - lf^2*q1d^2*cos(q2)*cos(q4)^2*sin(q3) \\
& + lf^2*q2d^2*cos(q2)*cos(q3)^2*sin(q3) - \\
& lf^2*q2d^2*cos(q2)*cos(q3)^2*cos(q4)^2*sin(q3) \\
& - 2*lf*lu*q1d*q4d*cos(q2)^2*cos(q4) + \\
& lf^2*q1d^2*cos(q2)^3*cos(q3)^2*cos(q4)^2*sin(q3) \\
& + 2*lf^2*q2d*q4d*cos(q2)*cos(q3)*sin(q3) - \\
& 2*lf^2*q1d*q2d*cos(q2)^2*cos(q3)^3*cos(q4)^2 \\
& - lf^2*q1d*q4d*cos(q2)^2*cos(q3)^2*cos(q4)^2 \\
& + lf^2*q1d*q3d*cos(q2)*sin(q2)*sin(q3) + \\
& 2*lf*lu*q1d^2*cos(q2)^3*cos(q4)*sin(q3) + \\
& lf^2*q2d*q3d*cos(q3)*cos(q4)^2*sin(q2) + \\
& lf*lu*q2d*q3d*cos(q2)*cos(q3)^2*sin(q4) + \\
& lf^2*q2d^2*cos(q3)*cos(q4)*sin(q2)*sin(q3)*sin(q4) \\
& + lf*lu*q1d*q4d*cos(q2)^2*cos(q3)^2*cos(q4) + \\
& lf^2*q3d*q4d*cos(q2)*cos(q3)*cos(q4)*sin(q4) \\
& - lf^2*q1d*q2d*cos(q2)*cos(q4)*sin(q2)*sin(q4) \\
& + lf*lu*q2d^2*cos(q3)*sin(q2)*sin(q3)*sin(q4) \\
& + lf^2*q2d*q4d*cos(q4)*sin(q2)*sin(q3)*sin(q4) \\
& - lf*lu*q1d*q2d*cos(q2)*sin(q2)*sin(q4) \\
& + lf*lu*q2d*q4d*sin(q2)*sin(q3)*sin(q4) + \\
& lf^2*q2d*q3d*cos(q2)*cos(q3)^2*cos(q4)*sin(q4) \\
& - lf^2*q2d*q4d*cos(q2)*cos(q3)*cos(q4)^2*sin(q3) \\
& - lf^2*q1d*q3d*cos(q2)*cos(q4)^2*sin(q2)*sin(q3) \\
& - 2*lf*lu*q1d*q2d*cos(q2)^2*cos(q3)*cos(q4) + \\
& 2*lf^2*q1d*q4d*cos(q2)*cos(q3)*cos(q4)*sin(q2)*sin(q4) \\
& + lf*lu*q2d*q4d*cos(q2)*cos(q3)*cos(q4)*sin(q3) - \\
& 2*lf^2*q1d^2*cos(q2)^2*cos(q3)*cos(q4)*sin(q2)*sin(q3)*sin(q4) \\
& + lf*lu*q1d*q4d*cos(q2)*cos(q3)*sin(q2)*sin(q4) + \\
& 3*lf^2*q1d*q2d*cos(q2)*cos(q3)^2*cos(q4)*sin(q2)*sin(q4) \\
& - lf^2*q1d*q3d*cos(q2)^2*cos(q3)*cos(q4)*sin(q3)*sin(q4) \\
& - 2*lf*lu*q1d^2*cos(q2)^2*cos(q3)*sin(q2)*sin(q3)*sin(q4) \\
& + 3*lf*lu*q1d*q2d*cos(q2)*cos(q3)^2*sin(q2)*sin(q4) \\
& - lf*lu*q1d*q3d*cos(q2)^2*cos(q3)*sin(q3)*sin(q4)))/ \\
& ((lu*q2d*cos(q2) - lf*q4d*sin(q2)*sin(q4) + \\
& lf*q2d*cos(q2)*cos(q4) + lf*q4d*cos(q2)*cos(q3)*cos(q4) - \\
& lf*q2d*cos(q3)*sin(q2)*sin(q4) - lf*q3d*cos(q2)*sin(q3)*sin(q4))^2 \\
& + (lu*q1d*cos(q2)*sin(q1) + lu*q2d*cos(q1)*sin(q2) + \\
& lf*q1d*cos(q2)*cos(q4)*sin(q1) + lf*q2d*cos(q1)*cos(q4)*sin(q2) \\
& + lf*q4d*cos(q1)*cos(q2)*sin(q4) - lf*q1d*cos(q1)*sin(q3)*sin(q4) \\
& - lf*q3d*cos(q3)*sin(q1)*sin(q4) - lf*q4d*cos(q4)*sin(q1)*sin(q3) \\
& + lf*q2d*cos(q1)*cos(q2)*cos(q3)*sin(q4) \\
& + lf*q4d*cos(q1)*cos(q3)*cos(q4)*sin(q2) \\
& - lf*q1d*cos(q3)*sin(q1)*sin(q2)*sin(q4) - \\
& lf*q3d*cos(q1)*sin(q2)*sin(q3)*sin(q4))^2 + \\
& (lu*q2d*sin(q1)*sin(q2) - lu*q1d*cos(q1)*cos(q2) - \\
& lf*q1d*cos(q1)*cos(q2)*cos(q4) + lf*q3d*cos(q1)*cos(q3)*sin(q4) \\
& + lf*q4d*cos(q1)*cos(q4)*sin(q3) + lf*q2d*cos(q4)*sin(q1)*sin(q2) \\
& + lf*q4d*cos(q2)*sin(q1)*sin(q4) - lf*q1d*sin(q1)*sin(q3)*sin(q4) \\
& + lf*q1d*cos(q1)*cos(q3)*sin(q2)*sin(q4) \\
& + lf*q2d*cos(q2)*cos(q3)*sin(q1)*sin(q4) \\
& + lf*q4d*cos(q3)*cos(q4)*sin(q1)*sin(q2) - \\
& lf*q3d*sin(q1)*sin(q2)*sin(q3)*sin(q4))^2) - \\
& Fn*sin(q1)*(((lu*q2d*cos(q2) - lf*q4d*sin(q2)*sin(q4) +
\end{aligned}$$

$$\begin{aligned}
& lf*q2d*cos(q2)*cos(q4) + lf*q4d*cos(q2)*cos(q3)*cos(q4) - \\
& lf*q2d*cos(q3)*sin(q2)*sin(q4) - \\
& lf*q3d*cos(q2)*sin(q3)*sin(q4))*(lf*q4d*cos(q1)*cos(q2)*cos(q4) - \\
& lu*q1d*cos(q1)*cos(q2)^2*sin(q3) + lu*q2d*cos(q1)*cos(q2)*cos(q3) \\
& - lf*q1d*cos(q2)*sin(q1)*sin(q4) + lf*q4d*sin(q1)*sin(q3)*sin(q4) \\
& + lf*q1d*cos(q2)*cos(q3)^2*sin(q1)*sin(q4) \\
& - lf*q2d*cos(q1)*cos(q3)^2*sin(q2)*sin(q4) \\
& + lf*q2d*cos(q1)*cos(q2)*cos(q3)*cos(q4) \\
& - lf*q4d*cos(q1)*cos(q3)*sin(q2)*sin(q4) \\
& + lf*q2d*cos(q3)*sin(q1)*sin(q3)*sin(q4) - \\
& lf*q1d*cos(q1)*cos(q2)^2*cos(q4)*sin(q3) + \\
& lf*q1d*cos(q1)*cos(q2)*cos(q3)*sin(q2)*sin(q3)*sin(q4)))/ \\
& ((lu*q2d*cos(q2) - lf*q4d*sin(q2)*sin(q4) + lf*q2d*cos(q2)*cos(q4) \\
& + lf*q4d*cos(q2)*cos(q3)*cos(q4) - lf*q2d*cos(q3)*sin(q2)*sin(q4) \\
& - lf*q3d*cos(q2)*sin(q3)*sin(q4))^2 + (q4d*(lf*(cos(q4)*sin(q2) + \\
& cos(q2)*cos(q3)*sin(q4))*(cos(q1)*cos(q3) - sin(q1)*sin(q2)*sin(q3)) \\
& + lf*cos(q2)*sin(q3)*(cos(q1)*sin(q3)*sin(q4) - \\
& cos(q2)*cos(q4)*sin(q1) + cos(q3)*sin(q1)*sin(q2)*sin(q4))) \\
& + q1d*(lu*cos(q2)*sin(q1) + lf*cos(q2)*cos(q4)*sin(q1) - \\
& lf*cos(q1)*sin(q3)*sin(q4) - lf*cos(q3)*sin(q1)*sin(q2)*sin(q4)) \\
& + q2d*cos(q1)*(lu*sin(q2) + lf*cos(q4)*sin(q2) + \\
& lf*cos(q2)*cos(q3)*sin(q4)) - lf*q3d*sin(q4)*(cos(q3)*sin(q1) + \\
& cos(q1)*sin(q2)*sin(q3)))^2 + (lf*q4d*(cos(q1)*cos(q4)*sin(q3) \\
& + cos(q2)*sin(q1)*sin(q4) + cos(q3)*cos(q4)*sin(q1)*sin(q2)) \\
& - q1d*(lu*cos(q1)*cos(q2) + lf*cos(q1)*cos(q2)*cos(q4) + \\
& lf*sin(q1)*sin(q3)*sin(q4) - lf*cos(q1)*cos(q3)*sin(q2)*sin(q4)) \\
& + q2d*sin(q1)*(lu*sin(q2) + lf*cos(q4)*sin(q2) + \\
& lf*cos(q2)*cos(q3)*sin(q4)) + lf*q3d*sin(q4)*(cos(q1)*cos(q3) \\
& - sin(q1)*sin(q2)*sin(q3)))^2) + ((q4d*(lf*(cos(q4)*sin(q2) + \\
& cos(q2)*cos(q3)*sin(q4))*(cos(q1)*cos(q3) - sin(q1)*sin(q2)*sin(q3)) \\
& + lf*cos(q2)*sin(q3)*(cos(q1)*sin(q3)*sin(q4) - \\
& cos(q2)*cos(q4)*sin(q1) + cos(q3)*sin(q1)*sin(q2)*sin(q4))) \\
& + q1d*(lu*cos(q2)*sin(q1) + lf*cos(q2)*cos(q4)*sin(q1) - \\
& lf*cos(q1)*sin(q3)*sin(q4) - lf*cos(q3)*sin(q1)*sin(q2)*sin(q4)) \\
& + q2d*cos(q1)*(lu*sin(q2) + lf*cos(q4)*sin(q2) + \\
& lf*cos(q2)*cos(q3)*sin(q4)) - lf*q3d*sin(q4)*(cos(q3)*sin(q1) \\
& + cos(q1)*sin(q2)*sin(q3)))*(lf*q4d*cos(q4)*sin(q2) + \\
& lu*q2d*cos(q3)*sin(q2) + lf*q2d*cos(q2)*cos(q3)^2*sin(q4) + \\
& lf*q2d*cos(q3)*cos(q4)*sin(q2) + lf*q4d*cos(q2)*cos(q3)*sin(q4) - \\
& lu*q1d*cos(q2)*sin(q2)*sin(q3) - \\
& lf*q1d*cos(q2)^2*cos(q3)*sin(q3)*sin(q4) - \\
& lf*q1d*cos(q2)*cos(q4)*sin(q2)*sin(q3)))/((lu*q2d*cos(q2) \\
& - lf*q4d*sin(q2)*sin(q4) + lf*q2d*cos(q2)*cos(q4) + \\
& lf*q4d*cos(q2)*cos(q3)*cos(q4) - lf*q2d*cos(q3)*sin(q2)*sin(q4) \\
& - lf*q3d*cos(q2)*sin(q3)*sin(q4))^2 + (q4d*(lf*(cos(q4)*sin(q2) + \\
& cos(q2)*cos(q3)*sin(q4))*(cos(q1)*cos(q3) - sin(q1)*sin(q2)*sin(q3)) \\
& + lf*cos(q2)*sin(q3)*(cos(q1)*sin(q3)*sin(q4) - \\
& cos(q2)*cos(q4)*sin(q1) + cos(q3)*sin(q1)*sin(q2)*sin(q4))) \\
& + q1d*(lu*cos(q2)*sin(q1) + lf*cos(q2)*cos(q4)*sin(q1) - \\
& lf*cos(q1)*sin(q3)*sin(q4) - lf*cos(q3)*sin(q1)*sin(q2)*sin(q4)) \\
& + q2d*cos(q1)*(lu*sin(q2) + lf*cos(q4)*sin(q2) + \\
& lf*cos(q2)*cos(q3)*sin(q4)) - lf*q3d*sin(q4)*(cos(q3)*sin(q1) + \\
& cos(q1)*sin(q2)*sin(q3)))^2 + (lf*q4d*(cos(q1)*cos(q4)*sin(q3) \\
& + cos(q2)*sin(q1)*sin(q4) + cos(q3)*cos(q4)*sin(q1)*sin(q2))
\end{aligned}$$

$$\begin{aligned}
& - q1d*(lu*\cos(q1)*\cos(q2) + lf*\cos(q1)*\cos(q2)*\cos(q4) + \\
& lf*\sin(q1)*\sin(q3)*\sin(q4) - lf*\cos(q1)*\cos(q3)*\sin(q2)*\sin(q4)) \\
& + q2d*\sin(q1)*(lu*\sin(q2) + lf*\cos(q4)*\sin(q2) + \\
& lf*\cos(q2)*\cos(q3)*\sin(q4)) + lf*q3d*\sin(q4)*(\cos(q1)*\cos(q3) - \\
& \sin(q1)*\sin(q2)*\sin(q3)))^2)*(lu*\sin(q2) + lf*\cos(q4)*\sin(q2) + \\
& lf*\cos(q2)*\cos(q3)*\sin(q4))
\end{aligned}$$

$$\begin{aligned}
& -(Fn*lf*\sin(q4)*(lf^2*q4d^2 + lf^2*q1d^2*\cos(q2)^2 \\
& + lf^2*q2d^2*\cos(q3)^2 + lu^2*q1d^2*\cos(q2)^2 + lu^2*q2d^2*\cos(q3)^2 \\
& - lf^2*q1d^2*\cos(q2)^2*\cos(q3)^2 - lu^2*q1d^2*\cos(q2)^2*\cos(q3)^2 \\
& + 2*lf^2*q2d*q4d*\cos(q3) + 2*lf*lu*q1d^2*\cos(q2)^2*\cos(q4) + \\
& 2*lf*lu*q2d^2*\cos(q3)^2*\cos(q4) - 2*lf^2*q1d*q4d*\cos(q2)*\sin(q3) \\
& - 2*lf*lu*q1d^2*\cos(q2)^2*\cos(q3)^2*\cos(q4) \\
& - 2*lf^2*q1d*q2d*\cos(q2)*\cos(q3)*\sin(q3) \\
& - 2*lu^2*q1d*q2d*\cos(q2)*\cos(q3)*\sin(q3) + \\
& 2*lf*lu*q2d*q4d*\cos(q3)*\cos(q4) - \\
& 2*lf*lu*q1d*q4d*\cos(q2)*\cos(q4)*\sin(q3) - \\
& 4*lf*lu*q1d*q2d*\cos(q2)*\cos(q3)*\cos(q4)*\sin(q3)))/ \\
& ((lu*q2d*\cos(q2) - lf*q4d*\sin(q2)*\sin(q4) + \\
& lf*q2d*\cos(q2)*\cos(q4) + lf*q4d*\cos(q2)*\cos(q3)*\cos(q4) - \\
& lf*q2d*\cos(q3)*\sin(q2)*\sin(q4) - lf*q3d*\cos(q2)*\sin(q3)*\sin(q4))^2 \\
& + (lu*q1d*\cos(q2)*\sin(q1) + lu*q2d*\cos(q1)*\sin(q2) + \\
& lf*q1d*\cos(q2)*\cos(q4)*\sin(q1) + lf*q2d*\cos(q1)*\cos(q4)*\sin(q2) \\
& + lf*q4d*\cos(q1)*\cos(q2)*\sin(q4) - lf*q1d*\cos(q1)*\sin(q3)*\sin(q4) \\
& - lf*q3d*\cos(q3)*\sin(q1)*\sin(q4) - lf*q4d*\cos(q4)*\sin(q1)*\sin(q3) \\
& + lf*q2d*\cos(q1)*\cos(q2)*\cos(q3)*\sin(q4) \\
& + lf*q4d*\cos(q1)*\cos(q3)*\cos(q4)*\sin(q2) \\
& - lf*q1d*\cos(q3)*\sin(q1)*\sin(q2)*\sin(q4) - \\
& lf*q3d*\cos(q1)*\sin(q2)*\sin(q3)*\sin(q4))^2 + \\
& (lu*q2d*\sin(q1)*\sin(q2) - lu*q1d*\cos(q1)*\cos(q2) - \\
& lf*q1d*\cos(q1)*\cos(q2)*\cos(q4) + lf*q3d*\cos(q1)*\cos(q3)*\sin(q4) \\
& + lf*q4d*\cos(q1)*\cos(q4)*\sin(q3) + lf*q2d*\cos(q4)*\sin(q1)*\sin(q2) \\
& + lf*q4d*\cos(q2)*\sin(q1)*\sin(q4) - lf*q1d*\sin(q1)*\sin(q3)*\sin(q4) \\
& + lf*q1d*\cos(q1)*\cos(q3)*\sin(q2)*\sin(q4) \\
& + lf*q2d*\cos(q2)*\cos(q3)*\sin(q1)*\sin(q4) \\
& + lf*q4d*\cos(q3)*\cos(q4)*\sin(q1)*\sin(q2) - \\
& lf*q3d*\sin(q1)*\sin(q2)*\sin(q3)*\sin(q4))^2) \\
& Fn*(((lf*q4d*(\cos(q1)*\cos(q4)*\sin(q3) + \cos(q2)*\sin(q1)*\sin(q4) \\
& + \cos(q3)*\cos(q4)*\sin(q1)*\sin(q2)) - q1d*(lu*\cos(q1)*\cos(q2) \\
& + lf*\cos(q1)*\cos(q2)*\cos(q4) + lf*\sin(q1)*\sin(q3)*\sin(q4) \\
& - lf*\cos(q1)*\cos(q3)*\sin(q2)*\sin(q4)) +
\end{aligned}$$

$$\begin{aligned}
& q2d*\sin(q1)*(lu*\sin(q2) + lf*\cos(q4)*\sin(q2) + \\
& lf*\cos(q2)*\cos(q3)*\sin(q4)) + lf*q3d*\sin(q4)*(cos(q1)*cos(q3) \\
& - \sin(q1)*\sin(q2)*\sin(q3)))*(lf*q4d*\cos(q4)*\sin(q2) + \\
& lu*q2d*\cos(q3)*\sin(q2) + lf*q2d*\cos(q2)*\cos(q3)^2*\sin(q4) + \\
& lf*q2d*\cos(q3)*\cos(q4)*\sin(q2) + lf*q4d*\cos(q2)*\cos(q3)*\sin(q4) - \\
& lu*q1d*\cos(q2)*\sin(q2)*\sin(q3) - \\
& lf*q1d*\cos(q2)^2*\cos(q3)*\sin(q3)*\sin(q4) - \\
& lf*q1d*\cos(q2)*\cos(q4)*\sin(q2)*\sin(q3))) / ((lu*q2d*\cos(q2) \\
& - lf*q4d*\sin(q2)*\sin(q4) + lf*q2d*\cos(q2)*\cos(q4) + \\
& lf*q4d*\cos(q2)*\cos(q3)*\cos(q4) - lf*q2d*\cos(q3)*\sin(q2)*\sin(q4) \\
& - lf*q3d*\cos(q2)*\sin(q3)*\sin(q4))^2 + (q4d*(lf*(cos(q4)*\sin(q2) + \\
& cos(q2)*\cos(q3)*\sin(q4))*(cos(q1)*cos(q3) - \sin(q1)*\sin(q2)*\sin(q3)) \\
& + lf*\cos(q2)*\sin(q3)*(cos(q1)*\sin(q3)*\sin(q4) - \\
& cos(q2)*\cos(q4)*\sin(q1) + cos(q3)*\sin(q1)*\sin(q2)*\sin(q4))) \\
& + q1d*(lu*\cos(q2)*\sin(q1) + lf*\cos(q2)*\cos(q4)*\sin(q1) - \\
& lf*\cos(q1)*\sin(q3)*\sin(q4) - lf*\cos(q3)*\sin(q1)*\sin(q2)*\sin(q4)) \\
& + q2d*\cos(q1)*(lu*\sin(q2) + lf*\cos(q4)*\sin(q2) + \\
& lf*\cos(q2)*\cos(q3)*\sin(q4)) - lf*q3d*\sin(q4)*(cos(q3)*\sin(q1) + \\
& cos(q1)*\sin(q2)*\sin(q3)))^2 + (lf*q4d*(cos(q1)*cos(q4)*\sin(q3) \\
& + cos(q2)*\sin(q1)*\sin(q4) + cos(q3)*cos(q4)*\sin(q1)*\sin(q2)) \\
& - q1d*(lu*\cos(q1)*cos(q2) + lf*\cos(q1)*cos(q2)*cos(q4) + \\
& lf*\sin(q1)*\sin(q3)*\sin(q4) - lf*\cos(q1)*cos(q3)*\sin(q2)*\sin(q4)) \\
& + q2d*\sin(q1)*(lu*\sin(q2) + lf*\cos(q4)*\sin(q2) + \\
& lf*\cos(q2)*\cos(q3)*\sin(q4)) + lf*q3d*\sin(q4)*(cos(q1)*cos(q3) \\
& - \sin(q1)*\sin(q2)*\sin(q3)))^2) - ((lu*q2d*\cos(q2) \\
& - lf*q4d*\sin(q2)*\sin(q4) + lf*q2d*\cos(q2)*\cos(q4) + \\
& lf*q4d*\cos(q2)*\cos(q3)*\cos(q4) - lf*q2d*\cos(q3)*\sin(q2)*\sin(q4) - \\
& lf*q3d*\cos(q2)*\sin(q3)*\sin(q4))*(lu*q1d*\cos(q2)^2*\sin(q1)*\sin(q3) \\
& - lf*q1d*\cos(q1)*cos(q2)*\sin(q4) - lf*q4d*\cos(q2)*cos(q4)*\sin(q1) \\
& - lu*q2d*\cos(q2)*cos(q3)*\sin(q1) + lf*q4d*\cos(q1)*\sin(q3)*\sin(q4) \\
& + lf*q1d*\cos(q2)^2*\cos(q4)*\sin(q1)*\sin(q3) \\
& + lf*q2d*\cos(q3)^2*\sin(q1)*\sin(q2)*\sin(q4) \\
& - lf*q2d*\cos(q2)*cos(q3)*cos(q4)*\sin(q1) \\
& + lf*q2d*\cos(q1)*cos(q3)*\sin(q3)*\sin(q4) \\
& + lf*q4d*\cos(q3)*\sin(q1)*\sin(q2)*\sin(q4) + \\
& lf*q1d*\cos(q1)*cos(q2)*cos(q3)^2*\sin(q4) - \\
& lf*q1d*\cos(q2)*cos(q3)*\sin(q1)*\sin(q2)*\sin(q3)*\sin(q4))) / \\
& ((lu*q2d*\cos(q2) - lf*q4d*\sin(q2)*\sin(q4) + lf*q2d*\cos(q2)*\cos(q4) \\
& + lf*q4d*\cos(q2)*cos(q3)*cos(q4) - lf*q2d*\cos(q3)*\sin(q2)*\sin(q4) \\
& - lf*q3d*\cos(q2)*\sin(q3)*\sin(q4))^2 + (q4d*(lf*(cos(q4)*\sin(q2) + \\
& cos(q2)*cos(q3)*\sin(q4))*(cos(q1)*cos(q3) - \sin(q1)*\sin(q2)*\sin(q3)) \\
& + lf*\cos(q2)*\sin(q3)*(cos(q1)*\sin(q3)*\sin(q4) - \\
& cos(q2)*cos(q4)*\sin(q1) + cos(q3)*\sin(q1)*\sin(q2)*\sin(q4))) \\
& + q1d*(lu*\cos(q2)*\sin(q1) + lf*\cos(q2)*cos(q4)*\sin(q1) - \\
& lf*\cos(q1)*\sin(q3)*\sin(q4) - lf*\cos(q3)*\sin(q1)*\sin(q2)*\sin(q4)) \\
& + q2d*\cos(q1)*(lu*\sin(q2) + lf*\cos(q4)*\sin(q2) + \\
& lf*\cos(q2)*cos(q3)*\sin(q4)) - lf*q3d*\sin(q4)*(cos(q3)*\sin(q1) + \\
& cos(q1)*\sin(q2)*\sin(q3)))^2 + (lf*q4d*(cos(q1)*cos(q4)*\sin(q3) \\
& + cos(q2)*\sin(q1)*\sin(q4) + cos(q3)*cos(q4)*\sin(q1)*\sin(q2)) \\
& - q1d*(lu*\cos(q1)*cos(q2) + lf*\cos(q1)*cos(q2)*cos(q4) + \\
& lf*\sin(q1)*\sin(q3)*\sin(q4) - lf*\cos(q1)*cos(q3)*\sin(q2)*\sin(q4)) \\
& + q2d*\sin(q1)*(lu*\sin(q2) + lf*\cos(q4)*\sin(q2) + \\
& lf*\cos(q2)*cos(q3)*\sin(q4)) + lf*q3d*\sin(q4)*(cos(q1)*cos(q3) \\
& - \sin(q1)*\sin(q2)*\sin(q3)))^2))*(lf*(cos(q4)*\sin(q2) +
\end{aligned}$$

$$\begin{aligned}
& \cos(q_2) \cos(q_3) \sin(q_4)) (\cos(q_1) \cos(q_3) - \sin(q_1) \sin(q_2) \sin(q_3)) \\
& + lf \cos(q_2) \sin(q_3) (\cos(q_1) \sin(q_3) \sin(q_4) - \\
& \cos(q_2) \cos(q_4) \sin(q_1) + \cos(q_3) \sin(q_1) \sin(q_2) \sin(q_4))) \\
& - Fn lf (((lu q_2 d \cos(q_2) - lf q_4 d \sin(q_2) \sin(q_4) + \\
& lf q_2 d \cos(q_2) \cos(q_4) + lf q_4 d \cos(q_2) \cos(q_3) \cos(q_4) - \\
& lf q_2 d \cos(q_3) \sin(q_2) \sin(q_4) - \\
& lf q_3 d \cos(q_2) \sin(q_3) \sin(q_4)) (lf q_4 d \cos(q_1) \cos(q_2) \cos(q_4) - \\
& lu q_1 d \cos(q_1) \cos(q_2)^2 \sin(q_3) + lu q_2 d \cos(q_1) \cos(q_2) \cos(q_3) \\
& - lf q_1 d \cos(q_2) \sin(q_1) \sin(q_4) + lf q_4 d \sin(q_1) \sin(q_3) \sin(q_4) \\
& + lf q_1 d \cos(q_2) \cos(q_3)^2 \sin(q_1) \sin(q_4) \\
& - lf q_2 d \cos(q_1) \cos(q_3)^2 \sin(q_2) \sin(q_4) \\
& + lf q_2 d \cos(q_1) \cos(q_2) \cos(q_3) \cos(q_4) \\
& - lf q_4 d \cos(q_1) \cos(q_3) \sin(q_2) \sin(q_4) \\
& + lf q_2 d \cos(q_3) \sin(q_1) \sin(q_3) \sin(q_4) - \\
& lf q_1 d \cos(q_1) \cos(q_2)^2 \cos(q_4) \sin(q_3) + \\
& lf q_1 d \cos(q_1) \cos(q_2) \cos(q_3) \sin(q_2) \sin(q_3) \sin(q_4))) / \\
& ((lu q_2 d \cos(q_2) - lf q_4 d \sin(q_2) \sin(q_4) + lf q_2 d \cos(q_2) \cos(q_4) \\
& + lf q_4 d \cos(q_2) \cos(q_3) \cos(q_4) - lf q_2 d \cos(q_3) \sin(q_2) \sin(q_4) \\
& - lf q_3 d \cos(q_2) \sin(q_3) \sin(q_4))^2 + (q_4 d (lf (\cos(q_4) \sin(q_2) + \\
& \cos(q_2) \cos(q_3) \sin(q_4)) (\cos(q_1) \cos(q_3) - \sin(q_1) \sin(q_2) \sin(q_3)) \\
& + lf \cos(q_2) \sin(q_3) (\cos(q_1) \sin(q_3) \sin(q_4) - \\
& \cos(q_2) \cos(q_4) \sin(q_1) + \cos(q_3) \sin(q_1) \sin(q_2) \sin(q_4))) \\
& + q_1 d (lu \cos(q_2) \sin(q_1) + lf \cos(q_2) \cos(q_4) \sin(q_1) - \\
& lf \cos(q_1) \sin(q_3) \sin(q_4) - lf \cos(q_3) \sin(q_1) \sin(q_2) \sin(q_4)) \\
& + q_2 d \cos(q_1) (lu \sin(q_2) + lf \cos(q_4) \sin(q_2) + \\
& lf \cos(q_2) \cos(q_3) \sin(q_4)) - lf q_3 d \sin(q_4) (\cos(q_3) \sin(q_1) + \\
& \cos(q_1) \sin(q_2) \sin(q_3)))^2 + (lf q_4 d (\cos(q_1) \cos(q_4) \sin(q_3) \\
& + \cos(q_2) \sin(q_1) \sin(q_4) + \cos(q_3) \cos(q_4) \sin(q_1) \sin(q_2)) \\
& - q_1 d (lu \cos(q_1) \cos(q_2) + lf \cos(q_1) \cos(q_2) \cos(q_4) + \\
& lf \sin(q_1) \sin(q_3) \sin(q_4) - lf \cos(q_1) \cos(q_3) \sin(q_2) \sin(q_4)) \\
& + q_2 d \sin(q_1) (lu \sin(q_2) + lf \cos(q_4) \sin(q_2) + \\
& lf \cos(q_2) \cos(q_3) \sin(q_4)) + lf q_3 d \sin(q_4) (\cos(q_1) \cos(q_3) \\
& - \sin(q_1) \sin(q_2) \sin(q_3)))^2) + ((q_4 d (lf (\cos(q_4) \sin(q_2) + \\
& \cos(q_2) \cos(q_3) \sin(q_4)) (\cos(q_1) \cos(q_3) - \sin(q_1) \sin(q_2) \sin(q_3)) \\
& + lf \cos(q_2) \sin(q_3) (\cos(q_1) \sin(q_3) \sin(q_4) - \\
& \cos(q_2) \cos(q_4) \sin(q_1) + \cos(q_3) \sin(q_1) \sin(q_2) \sin(q_4))) \\
& + q_1 d (lu \cos(q_2) \sin(q_1) + lf \cos(q_2) \cos(q_4) \sin(q_1) - \\
& lf \cos(q_1) \sin(q_3) \sin(q_4) - lf \cos(q_3) \sin(q_1) \sin(q_2) \sin(q_4)) \\
& + q_2 d \cos(q_1) (lu \sin(q_2) + lf \cos(q_4) \sin(q_2) + \\
& lf \cos(q_2) \cos(q_3) \sin(q_4)) - lf q_3 d \sin(q_4) (\cos(q_3) \sin(q_1) \\
& + \cos(q_1) \sin(q_2) \sin(q_3))) (lf q_4 d \cos(q_4) \sin(q_2) + \\
& lu q_2 d \cos(q_3) \sin(q_2) + lf q_2 d \cos(q_2) \cos(q_3)^2 \sin(q_4) + \\
& lf q_2 d \cos(q_3) \cos(q_4) \sin(q_2) + lf q_4 d \cos(q_2) \cos(q_3) \sin(q_4) - \\
& lu q_1 d \cos(q_2) \sin(q_2) \sin(q_3) - \\
& lf q_1 d \cos(q_2)^2 \cos(q_3) \sin(q_3) \sin(q_4) - \\
& lf q_1 d \cos(q_2) \cos(q_4) \sin(q_2) \sin(q_3))) / ((lu q_2 d \cos(q_2) \\
& - lf q_4 d \sin(q_2) \sin(q_4) + lf q_2 d \cos(q_2) \cos(q_4) + \\
& lf q_4 d \cos(q_2) \cos(q_3) \cos(q_4) - lf q_2 d \cos(q_3) \sin(q_2) \sin(q_4) \\
& - lf q_3 d \cos(q_2) \sin(q_3) \sin(q_4))^2 + (q_4 d (lf (\cos(q_4) \sin(q_2) + \\
& \cos(q_2) \cos(q_3) \sin(q_4)) (\cos(q_1) \cos(q_3) - \sin(q_1) \sin(q_2) \sin(q_3)) \\
& + lf \cos(q_2) \sin(q_3) (\cos(q_1) \sin(q_3) \sin(q_4) - \\
& \cos(q_2) \cos(q_4) \sin(q_1) + \cos(q_3) \sin(q_1) \sin(q_2) \sin(q_4))) \\
& + q_1 d (lu \cos(q_2) \sin(q_1) + lf \cos(q_2) \cos(q_4) \sin(q_1) - \\
& lf \cos(q_1) \sin(q_3) \sin(q_4) - lf \cos(q_3) \sin(q_1) \sin(q_2) \sin(q_4))
\end{aligned}$$

$$\begin{aligned}
& + q2d*\cos(q1)*(lu*\sin(q2) + lf*\cos(q4)*\sin(q2) + \\
& lf*\cos(q2)*\cos(q3)*\sin(q4)) - lf*q3d*\sin(q4)*(cos(q3)*\sin(q1) + \\
& cos(q1)*\sin(q2)*\sin(q3)))^2 + (lf*q4d*(cos(q1)*\cos(q4)*\sin(q3) \\
& + cos(q2)*\sin(q1)*\sin(q4) + cos(q3)*\cos(q4)*\sin(q1)*\sin(q2)) \\
& - q1d*(lu*\cos(q1)*\cos(q2) + lf*\cos(q1)*\cos(q2)*\cos(q4) + \\
& lf*\sin(q1)*\sin(q3)*\sin(q4) - lf*\cos(q1)*\cos(q3)*\sin(q2)*\sin(q4)) \\
& + q2d*\sin(q1)*(lu*\sin(q2) + lf*\cos(q4)*\sin(q2) + \\
& lf*\cos(q2)*\cos(q3)*\sin(q4)) + lf*q3d*\sin(q4)*(cos(q1)*\cos(q3) \\
& - \sin(q1)*\sin(q2)*\sin(q3)))^2)*(cos(q1)*\cos(q4)*\sin(q3) + \\
& cos(q2)*\sin(q1)*\sin(q4) + cos(q3)*\cos(q4)*\sin(q1)*\sin(q2)) - \\
& (Fn*lf*(\sin(q2)*\sin(q4) - \\
& cos(q2)*\cos(q3)*\cos(q4))*(lu^2*q1d^2*\cos(q2)^3*\sin(q3) - \\
& lf^2*q1d*q4d - lf^2*q1d*q4d*\cos(q2)^2 + lf^2*q1d*q4d*\cos(q4)^2 \\
& + lf^2*q1d^2*\cos(q2)*\sin(q3) + lf^2*q4d^2*\cos(q2)*\sin(q3) \\
& - lf^2*q1d*q2d*\cos(q3) - lf^2*q3d*q4d*\sin(q2) - \\
& lf^2*q1d*q2d*\cos(q2)^2*\cos(q3) + lf^2*q1d*q2d*\cos(q3)*\cos(q4)^2 \\
& - lu^2*q1d*q2d*\cos(q2)^2*\cos(q3) + lf^2*q3d*q4d*\cos(q4)^2*\sin(q2) \\
& - lf^2*q1d^2*\cos(q2)^3*\cos(q3)^2*\sin(q3) \\
& + lf^2*q1d^2*\cos(q2)^3*\cos(q4)^2*\sin(q3) \\
& + 2*lf^2*q1d*q2d*\cos(q2)^2*\cos(q3)^3 + \\
& 2*lf^2*q1d*q4d*\cos(q2)^2*\cos(q3)^2 - lf^2*q1d*q4d*\cos(q2)^2*\cos(q4)^2 \\
& - lf^2*q2d*q3d*\cos(q3)*\sin(q2) - lf^2*q1d^2*\cos(q2)*\cos(q4)^2*\sin(q3) \\
& + lf^2*q2d^2*\cos(q2)*\cos(q3)^2*\sin(q3) - \\
& lf^2*q2d^2*\cos(q2)*\cos(q3)^2*\cos(q4)^2*\sin(q3) \\
& - 2*lf*lu*q1d*q4d*\cos(q2)^2*\cos(q4) + \\
& lf^2*q1d^2*\cos(q2)^3*\cos(q3)^2*\cos(q4)^2*\sin(q3) \\
& + 2*lf^2*q2d*q4d*\cos(q2)*\cos(q3)*\sin(q3) - \\
& 2*lf^2*q1d*q2d*\cos(q2)^2*\cos(q3)^3*\cos(q4)^2 \\
& - lf^2*q1d*q4d*\cos(q2)^2*\cos(q3)^2*\cos(q4)^2 \\
& + lf^2*q1d*q3d*\cos(q2)*\sin(q2)*\sin(q3) + \\
& 2*lf*lu*q1d^2*\cos(q2)^3*\cos(q4)*\sin(q3) + \\
& lf^2*q2d*q3d*\cos(q3)*\cos(q4)^2*\sin(q2) + \\
& lf*lu*q2d*q3d*\cos(q2)*\cos(q3)^2*\sin(q4) + \\
& lf^2*q2d^2*\cos(q3)*\cos(q4)*\sin(q2)*\sin(q3)*\sin(q4) \\
& + lf*lu*q1d*q4d*\cos(q2)^2*\cos(q3)^2*\cos(q4) + \\
& lf^2*q3d*q4d*\cos(q2)*\cos(q3)*\cos(q4)*\sin(q4) \\
& - lf^2*q1d*q2d*\cos(q2)*\cos(q4)*\sin(q2)*\sin(q4) \\
& + lf*lu*q2d^2*\cos(q3)*\sin(q2)*\sin(q3)*\sin(q4) \\
& + lf^2*q2d*q4d*\cos(q4)*\sin(q2)*\sin(q3)*\sin(q4) \\
& - lf*lu*q1d*q2d*\cos(q2)*\sin(q2)*\sin(q4) \\
& + lf*lu*q2d*q4d*\sin(q2)*\sin(q3)*\sin(q4) + \\
& lf^2*q2d*q3d*\cos(q2)*\cos(q3)^2*\cos(q4)*\sin(q4) \\
& - lf^2*q2d*q4d*\cos(q2)*\cos(q3)*\cos(q4)^2*\sin(q3) \\
& - lf^2*q1d*q3d*\cos(q2)*\cos(q4)^2*\sin(q2)*\sin(q3) \\
& - 2*lf*lu*q1d*q2d*\cos(q2)^2*\cos(q3)*\cos(q4) + \\
& 2*lf^2*q1d*q4d*\cos(q2)*\cos(q3)*\cos(q4)*\sin(q2)*\sin(q4) \\
& + lf*lu*q2d*q4d*\cos(q2)*\cos(q3)*\cos(q4)*\sin(q3) - \\
& 2*lf^2*q1d^2*\cos(q2)^2*\cos(q3)*\cos(q4)*\sin(q2)*\sin(q3)*\sin(q4) \\
& + lf*lu*q1d*q4d*\cos(q2)*\cos(q3)*\sin(q2)*\sin(q4) + \\
& 3*lf^2*q1d*q2d*\cos(q2)*\cos(q3)^2*\cos(q4)*\sin(q2)*\sin(q4) \\
& - lf^2*q1d*q3d*\cos(q2)^2*\cos(q3)*\cos(q4)*\sin(q3)*\sin(q4) \\
& - 2*lf*lu*q1d^2*\cos(q2)^2*\cos(q3)*\sin(q2)*\sin(q3)*\sin(q4) \\
& + 3*lf*lu*q1d*q2d*\cos(q2)*\cos(q3)^2*\sin(q2)*\sin(q4) \\
& - lf*lu*q1d*q3d*\cos(q2)^2*\cos(q3)*\sin(q3)*\sin(q4)))/
\end{aligned}$$

$$\begin{aligned}
& ((lu*q2d*cos(q2) - lf*q4d*sin(q2)*sin(q4) + \\
& lf*q2d*cos(q2)*cos(q4) + lf*q4d*cos(q2)*cos(q3)*cos(q4) - \\
& lf*q2d*cos(q3)*sin(q2)*sin(q4) - lf*q3d*cos(q2)*sin(q3)*sin(q4))^2 \\
& + (lu*q1d*cos(q2)*sin(q1) + lu*q2d*cos(q1)*sin(q2) + \\
& lf*q1d*cos(q2)*cos(q4)*sin(q1) + lf*q2d*cos(q1)*cos(q4)*sin(q2) \\
& + lf*q4d*cos(q1)*cos(q2)*sin(q4) - lf*q1d*cos(q1)*sin(q3)*sin(q4) \\
& - lf*q3d*cos(q3)*sin(q1)*sin(q4) - lf*q4d*cos(q4)*sin(q1)*sin(q3) \\
& + lf*q2d*cos(q1)*cos(q2)*cos(q3)*sin(q4) \\
& + lf*q4d*cos(q1)*cos(q3)*cos(q4)*sin(q2) \\
& - lf*q1d*cos(q3)*sin(q1)*sin(q2)*sin(q4) - \\
& lf*q3d*cos(q1)*sin(q2)*sin(q3)*sin(q4))^2 + \\
& (lu*q2d*sin(q1)*sin(q2) - lu*q1d*cos(q1)*cos(q2) - \\
& lf*q1d*cos(q1)*cos(q2)*cos(q4) + lf*q3d*cos(q1)*cos(q3)*sin(q4) \\
& + lf*q4d*cos(q1)*cos(q4)*sin(q3) + lf*q2d*cos(q4)*sin(q1)*sin(q2) \\
& + lf*q4d*cos(q2)*sin(q1)*sin(q4) - lf*q1d*sin(q1)*sin(q3)*sin(q4) \\
& + lf*q1d*cos(q1)*cos(q3)*sin(q2)*sin(q4) \\
& + lf*q2d*cos(q2)*cos(q3)*sin(q1)*sin(q4) \\
& + lf*q4d*cos(q3)*cos(q4)*sin(q1)*sin(q2) - \\
& lf*q3d*sin(q1)*sin(q2)*sin(q3)*sin(q4))^2)
\end{aligned}$$

tangentialTorqueSanityTest =

$$\begin{aligned}
& \frac{(Ft*lf*lu*q4d*sin(q4))}{(lf^2*q2d^2 + 2*lf^2*q2d*q4d + lf^2*q4d^2 + 2*cos(q4)*lf*lu*q2d^2 + 2*cos(q4)*lf*lu*q2d*q4d + lu^2*q2d^2)^{(1/2)}} \\
& - \frac{(Ft*lf*lu*q2d*sin(q4))}{(lf^2*q2d^2 + 2*lf^2*q2d*q4d + lf^2*q4d^2 + 2*cos(q4)*lf*lu*q2d^2 + 2*cos(q4)*lf*lu*q2d*q4d + lu^2*q2d^2)^{(1/2)}}
\end{aligned}$$

normalTorqueSanityTest =

$$\begin{aligned}
& Fn*(lf*cos(q2 + q4) + lu*cos(q2)) \\
& - Fn*lf*sin(q4)
\end{aligned}$$

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